Multipole Expansion: Five Charge Distributions

Comprehensive Worked Answer Key — Revised for Mathematical Clarity

Problem-solving thread

Each activity follows the same route: identify the source, locate $R_{\max}$, choose and normalize dQ, evaluate Q_n, apply symmetry, and interpret the exterior potential. The mathematical explanations below accompany every question.

Convention

The guide uses θ for polar angle and ϕ for azimuth, opposite the student's preferred naming. All expressions retain the lecture convention.

Activity 1: Choose the correct multipole setup

Physical starting point

Every example begins with source geometry, the charge element, and the convergence region. The index n labels angular patterns, not individual charges.

1a.

Explain the meaning of r, r', θ', dQ, Q_n, and $R_{\max}$. Write the moment formula and the first three terms of the potential.

Physical setup and step-by-step solution

Identify what moves and what stays fixed

The observation point (r, θ) is fixed while we sum over the source positions (r', θ', ϕ'). A source element carries actual charge dQ; the moment Q_n weights that charge by its position and angular pattern.

Write the moment before the potential

$$\boxed{Q_n = \int_{\text{source}} r'^n P_n(\cos\theta')\, dQ.}$$

Insert one moment at a time

$$V(r,\theta) = \frac{1}{4\pi\epsilon_0}\sum_{n=0}^{\infty}\frac{Q_n}{r^{n+1}}P_n(\cos\theta)$$

$$= \frac{1}{4\pi\epsilon_0}\left[\frac{Q_0}{r} + \frac{Q_1}{r^2}P_1(\cos\theta) + \frac{Q_2}{r^3}P_2(\cos\theta) + \cdots\right].$$

State the domain

This exterior expansion requires $r > R_{\max}$, where $R_{\max}$ is the greatest source distance from the chosen origin.

Keep the thread

Distinguish a small piece of physical charge dQ from its weighted moment Q_n.

1b.

Match each source to its charge element: point charges, a line, a ring, and a spherical cloud. Explain how to check normalization.

Physical setup and reasoning

For points, $Q_n = \sum_i q_i r_i'^n P_n(\cos\theta_i')$. For a line, $dQ = \lambda\, d\ell'$; for a ring, $dQ = \lambda a\, d\phi'$; for a cloud, $dQ = \rho r'^2 \sin\theta' dr' d\theta' d\phi'$. Always integrate dQ alone first: $\int dQ = q_{\text{total}} = Q_0$.

Keep the thread

For a ring $d\ell' = a\, d\phi'$; for a cloud $d\tau' = r'^2 \sin\theta' dr' d\theta' d\phi'$.

1c.

Predict which moments vanish for a distribution unchanged under $z \mapsto -z$, and for one whose signed charge reverses under that reflection. Justify using $P_n(-x) = (-1)^n P_n(x)$.

Physical setup and reasoning

For a reflection-even distribution, paired contributions are proportional to $P_n(x)+P_n(-x) = [1+(-1)^n]P_n(x)$, zero for odd n. For a reflection-odd signed distribution, contributions are proportional to $P_n(x) - P_n(-x) = [1-(-1)^n]P_n(x)$, zero for even n.

Keep the thread

Pair source elements at opposite z with the same r'.

Activity 2: A single point charge at the origin

Physical starting point

Begin with the simplest source. This example also reveals why the multipole moments depend on the chosen origin.

2a.

For a point charge $+q$ at the origin, identify $R_{\max}$ and evaluate Q_0 without a volume integral.

Physical setup and reasoning

All source charge is at $r' = 0$, so $R_{\max} = 0$. The monopole is $Q_0 = q(r')^0 P_0 = q$. The origin has no preferred direction.

Keep the thread

Use the discrete-charge sum and $P_0 = 1$.

2b.

Explain why every Q_n with $n \geq 1$ vanishes, even though the polar angle at the origin is undefined.

Physical setup and reasoning

Because $|P_n(\cos\theta')| \leq 1$, $|r'^n P_n| \leq r'^n \to 0$ as $r' \to 0$ for $n \geq 1$. Thus $Q_n = 0$ for every higher moment, independent of the undefined direction at the origin.

Keep the thread

The factor $r'^n P_n(\cos\theta')$ tends to zero as $r' \to 0$ for $n \geq 1$.

2c.

Write the complete exterior potential and check its angular dependence and domain. What changes if the coordinate origin moves?

Physical setup and reasoning

$$\boxed{V(r) = \frac{q}{4\pi\epsilon_0 r}}, \qquad r > 0.$$

It is angle-independent, as spherical symmetry demands. If the origin is moved, the charge acquires nonzero source radius r' and generally higher moments about that new origin.

Keep the thread

Only Q_0 survives for the centered charge.

Activity 3: Two opposite charges on the polar axis

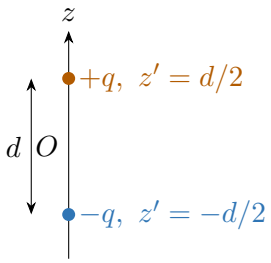
Physical starting point

For the axial dipole, the charge signs and position signs both matter. Calculate the first moments explicitly before generalizing.

3a.

Draw $+q$ at $z = d/2$ and $-q$ at $z = -d/2$. Label their radii and polar angles, then determine $R_{\max}$.

Physical setup and reasoning



Both charges have $r' = d/2$. The upper charge has $\theta' = 0$; the lower has $\theta' = \pi$. Hence $R_{\max} = d/2$.

Keep the thread

Both source radii equal $d/2$, but their polar angles are 0 and π.

3b.

Calculate Q_0, Q_1, Q_2, and Q_3 separately, showing the contribution from each charge. Explain the parity pattern.

Physical setup and step-by-step solution

Write the two source contributions separately

The upper source has charge $+q$, radius $d/2$, and $\cos\theta' = 1$. The lower source has charge $-q$, the same radius, and $\cos\theta' = -1$.

$$Q_n = (+q)\left(\frac{d}{2}\right)^n P_n(1) + (-q)\left(\frac{d}{2}\right)^n P_n(-1)$$

$$= q\left(\frac{d}{2}\right)^n [P_n(1) - P_n(-1)]$$

$$= q\left(\frac{d}{2}\right)^n [1 - (-1)^n]\,.$$

Monopole: $n = 0$

$$Q_0 = q - q$$

$$= 0.$$

Dipole: $n = 1$

$$Q_1 = (+q)\frac{d}{2} + (-q)\left(-\frac{d}{2}\right)$$

$$= \frac{qd}{2} + \frac{qd}{2}$$

$$= qd.$$

Quadrupole: $n = 2$

$$Q_2 = q\frac{d^2}{4} - q\frac{d^2}{4}$$

$$= 0.$$

Octupole: $n = 3$

$$Q_3 = q\left(\frac{d}{2}\right)^3\left[1 - (-1)^3\right]$$

$$= 2q\frac{d^3}{8}$$

$$= \frac{qd^3}{4}.$$

The source reverses sign under $z \mapsto -z$, so all even moments vanish.

Keep the thread

Use $P_n(1) = 1$ and $P_n(-1) = (-1)^n$.

3c.

Write the first two nonzero potential terms. Identify the leading far-field term and explain its direction.

Physical setup and reasoning

$$V(r,\theta) = \frac{1}{4\pi\epsilon_0}\left[\frac{qd\cos\theta}{r^2} + \frac{qd^3 P_3(\cos\theta)}{4r^4} + \cdots\right], \quad r > d/2.$$

For $r \gg d$, the dipole term dominates: $V \simeq qd\cos\theta/(4\pi\epsilon_0 r^2)$. The dipole moment is $qd\,\hat{\mathbf{z}}$.

Keep the thread

The dipole moment points from the negative charge to the positive charge.

Activity 4: A spherical charge cloud

Physical starting point

The density $f(r')$ can vary radially. Spherical symmetry, not uniform density, removes the higher exterior moments.

4a.

For $\rho = f(r')$ inside $r' \leq a$, write dQ, the integration limits, and the normalization equation for total charge q.

Physical setup and reasoning

$$dQ = f(r')r'^2 \sin\theta'\, dr' d\theta' d\phi'.$$

Limits: $0 \leq r' \leq a$, $0 \leq \theta' \leq \pi$, $0 \leq \phi' < 2\pi$. Integrating charge:

$$q = \int_0^a f(r')r'^2 dr' \int_0^\pi \sin\theta' d\theta' \int_0^{2\pi} d\phi' = 4\pi \int_0^a f(r')r'^2 dr'.$$

Keep the thread

The volume element contains $r'^2 \sin\theta'$.

4b.

Factor the moment integral into radial and angular pieces. With $u = \cos\theta'$, show why $Q_n = 0$ for $n \geq 1$.

Physical setup and step-by-step solution

Start from the correct volume charge element

$$dQ = f(r')r'^2 \sin\theta'\, dr'\, d\theta'\, d\phi'.$$

Insert it into the moment definition

$$Q_n = \int_0^a \int_0^\pi \int_0^{2\pi} r'^n P_n(\cos\theta') f(r')r'^2 \sin\theta'\, d\phi'\, d\theta'\, dr'.$$

Separate independent variables

$$Q_n = \left[\int_0^a f(r')r'^{n+2} dr'\right] \left[\int_0^\pi P_n(\cos\theta') \sin\theta'\, d\theta'\right] \left[\int_0^{2\pi} d\phi'\right].$$

The last integral equals 2π.

Change the polar integration variable

Let $u = \cos\theta'$, so $du = -\sin\theta'\,d\theta'$. The original limits $0, \pi$ become $1, -1$:

$$\int_0^{\pi} P_n(\cos\theta')\sin\theta'\,d\theta' = -\int_1^{-1} P_n(u)\,du$$

$$= \int_{-1}^{1} P_n(u)\,du$$

$$= 2\delta_{n0}.$$

The final equality uses orthogonality with $P_0(u) = 1$.

Interpret the cancellation

Thus $Q_n = 0$ for $n \geq 1$, while

$$Q_0 = 4\pi \int_0^a f(r')r'^2\,dr' = q.$$

No uniform-density assumption was needed.

Keep the thread

Show the transformed limits $\theta' = 0 \to u = 1$ and $\theta' = \pi \to u = -1$.

4c.

Give the exterior potential and explain why the same expression is not automatically valid inside the cloud.

Physical setup and reasoning

$$\boxed{V(r) = \frac{q}{4\pi\epsilon_0 r}}, \quad r > a.$$

Spherical symmetry makes the exterior potential identical to that of a point charge at the origin. Inside, the enclosed charge varies with radius and the potential depends on $f(r')$; the exterior expression cannot be extended inward without solving that region.

Keep the thread

The exterior series requires $r > a$; interior charge contributes differently.

Activity 5: A uniformly charged circular ring

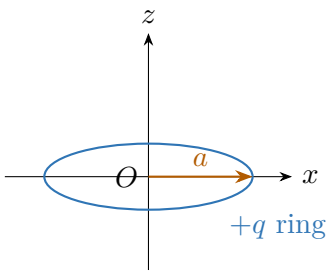
Physical starting point

All ring elements have the same radius and polar angle. Geometry makes the moment integral simple; the angular sign is important.

5a.

Sketch the ring in the xy plane. Label its radius a, polar angle θ', line density λ, and charge element dQ. Verify the total charge.

Physical setup and reasoning



Each source element has $r' = a$ and $\theta' = \pi/2$. The line density is $\lambda = q/(2\pi a)$, so $dQ = \lambda a\, d\phi' = q\, d\phi'/(2\pi)$. Integrating from 0 to 2π gives total charge q.

Keep the thread

Use $\theta' = \pi/2$, $d\ell' = a\, d\phi'$, and $\lambda = q/(2\pi a)$.

5b.

Derive $Q_n = qa^n P_n(0)$. Compute Q_0, Q_2, and Q_4 and explain why odd moments vanish.

Physical setup and step-by-step solution

Use the ring geometry before integrating

Every source element has $r' = a$ and $\cos\theta' = 0$. The charge element is

$$dQ = \frac{q}{2\pi}\, d\phi'.$$

Evaluate the angular integral

$$Q_n = \int_0^{2\pi} a^n P_n(0)\frac{q}{2\pi}\,d\phi'$$

$$= \frac{qa^n P_n(0)}{2\pi}\left[\phi'\right]_0^{2\pi}$$

$$= qa^n P_n(0).$$

Evaluate the first three even modes

$$Q_0 = qP_0(0) = q,$$

$$Q_2 = qa^2 P_2(0) = -\frac{qa^2}{2},$$

$$Q_4 = qa^4 P_4(0) = \frac{3qa^4}{8}.$$

Explain the missing odd modes

For odd n, $P_n(0) = 0$, so $Q_n = 0$. The negative Q_2 reflects the equatorial geometry, not negative charge.

Keep the thread

Use $P_2(0) = -1/2$ and $P_4(0) = 3/8$.

5c.

Write the first three nonzero terms of $V(r,\theta)$. Check the axis result against the exact ring potential $V(r,0) = q/(4\pi\epsilon_0\sqrt{r^2+a^2})$.

Physical setup and reasoning

$$V = \frac{q}{4\pi\epsilon_0 r}\left[1 - \frac{1}{2}\left(\frac{a}{r}\right)^2 P_2(\cos\theta) + \frac{3}{8}\left(\frac{a}{r}\right)^4 P_4(\cos\theta) + \cdots\right], \quad r > a.$$

On the axis $P_2(1) = P_4(1) = 1$, giving $1 - a^2/(2r^2) + 3a^4/(8r^4) + \cdots$. Independently, $1/\sqrt{r^2+a^2} = r^{-1}(1+a^2/r^2)^{-1/2}$ has the same binomial expansion.

Keep the thread

Factor out $1/r$ and expand $(1+a^2/r^2)^{-1/2}$.

Activity 6: A finite uniformly charged axial line

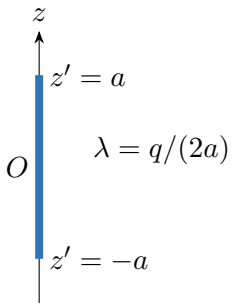
Physical starting point

Unlike the ring, the line lies on the polar axis. Its quadrupole has the opposite sign even though both carry positive charge.

6a.

Sketch the line from $z = -a$ to $z = a$. Find λ, $R_{\max}$, and the two polar angles needed to split the integral.

Physical setup and reasoning



The total length is $2a$, so $\lambda = q/(2a)$ and $R_{\max} = a$. For the upper half $P_n(\cos\theta') = P_n(1) = 1$; for the lower half $P_n(-1) = (-1)^n$.

Keep the thread

Upper half: $\theta' = 0$; lower half: $\theta' = \pi$.

6b.

Split the moment into upper and lower integrals over $0 \le r' \le a$. Integrate and obtain the general even/odd result.

Physical setup and step-by-step solution

Separate the upper and lower halves

The upper half has polar angle 0 and the lower half has polar angle π. Each half extends from source radius $r' = 0$ to $r' = a$.

$$Q_n = \lambda \int_0^a r'^n P_n(1)\, dr' + \lambda \int_0^a r'^n P_n(-1)\, dr'.$$

Use endpoint values of Legendre polynomials

$$Q_n = \lambda \int_0^a r'^n \, dr' + \lambda(-1)^n \int_0^a r'^n \, dr'$$

$$= \lambda \left[1 + (-1)^n\right] \int_0^a r'^n \, dr'.$$

Perform the power-law integral

$$\int_0^a r'^n \, dr' = \left[\frac{r'^{n+1}}{n+1}\right]_0^a$$

$$= \frac{a^{n+1}}{n+1}.$$

Therefore

$$Q_n = \frac{\lambda a^{n+1}}{n+1} \left[1 + (-1)^n\right].$$

Substitute the normalized line density

With $\lambda = q/(2a)$,

$$Q_n = \frac{qa^n}{2(n+1)} \left[1 + (-1)^n\right]$$

$$= \begin{cases} \dfrac{qa^n}{n+1}, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

The odd modes cancel between opposite halves of the same positively charged line.

Keep the thread

Use $\int_0^a r'^n dr' = a^{n+1}/(n+1)$.

6c.

Find Q_0, Q_2, and Q_4, write the exterior series through $n = 4$, and compare the sign of Q_2 with the ring.

Physical setup and reasoning

$Q_0 = q$, $Q_2 = qa^2/3$, and $Q_4 = qa^4/5$. Therefore

$$V = \frac{q}{4\pi\epsilon_0 r} \left[1 + \frac{1}{3}\left(\frac{a}{r}\right)^2 P_2(\cos\theta) + \frac{1}{5}\left(\frac{a}{r}\right)^4 P_4(\cos\theta) + \cdots\right], \quad r > a.$$

The line quadrupole is positive because $P_2(\pm 1) = 1$; the ring quadrupole is negative because $P_2(0) = -1/2$.

Keep the thread

On the polar axis $P_2(\pm 1) = 1$, whereas the ring has $P_2(0) = -1/2$.

Activity 7: Compare, combine, and choose the leading term

Physical starting point

Use symmetry and superposition to solve a new problem without starting the integration over again.

7a.

Complete a comparison: for the five lesson sources, give $R_{\max}$, which moments survive, and the leading far-field potential.

Physical setup and reasoning

Source	$R_{\max}$	Surviving modes	Leading V
Point at origin	0	$n = 0$	$q/(4\pi\epsilon_0 r)$
Axial $\pm q$	$d/2$	odd	$qd\cos\theta/(4\pi\epsilon_0 r^2)$
Spherical cloud	a	$n = 0$	$q/(4\pi\epsilon_0 r)$
Ring	a	even	$q/(4\pi\epsilon_0 r)$
Axial line	a	even	$q/(4\pi\epsilon_0 r)$

Keep the thread

For nonzero net charge, the monopole is leading even when higher moments are present.

7b.

A concentric spherical cloud carries $+q$ and a uniformly charged ring of radius a carries $-q$. Find Q_0 and Q_2 and the leading far-field potential.

Physical setup and step-by-step solution

Use linearity rather than reintegrating

The spherical cloud contributes only its monopole:

$$Q_0^{\text{cloud}} = +q, \qquad Q_2^{\text{cloud}} = 0.$$

The ring has charge $-q$, so its first moments are

$$Q_0^{\text{ring}} = -q,$$

$$Q_2^{\text{ring}} = (-q)a^2 P_2(0)$$

$$= (-q)a^2\left(-\frac{1}{2}\right)$$

$$= \frac{qa^2}{2}.$$

Add corresponding moments

$$Q_0^{\text{total}} = q - q = 0,$$

$$Q_1^{\text{total}} = 0,$$

$$Q_2^{\text{total}} = 0 + \frac{qa^2}{2} = \frac{qa^2}{2}.$$

The first nonzero mode is quadrupolar.

Insert it into the potential series

$$V(r,\theta) \simeq \frac{1}{4\pi\epsilon_0}\frac{Q_2}{r^3}P_2(\cos\theta)$$

$$= \boxed{\frac{qa^2}{8\pi\epsilon_0 r^3}P_2(\cos\theta)}.$$

This holds in the far field beyond both the ring and the cloud.

Keep the thread

The cloud has only a monopole; the ring has $Q_2 = q_{\text{ring}}a^2P_2(0)$.

7c.

For a generic source, state three checks that can catch an incorrect moment calculation: normalization, units, and convergence domain. Explain how each check works.

Physical setup and reasoning

Normalization: $Q_0 = \int dQ$ must equal net charge. Units: each factor r'^n gives $[Q_n] = \text{C}\,\text{m}^n$, so Q_n/r^{n+1} has units C/m and multiplication by $1/(4\pi\epsilon_0)$ gives volts. Domain: require $r > R_{\max}$ for every source point; merely being outside some portion of the material is insufficient.

Keep the thread

$[Q_n] = \text{C}\,\text{m}^n$ and the series needs $r > R_{\max}$.